Security-Aware Retrieval-Augmented Generation System for Post-Quantum Cryptography Intelligence

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***Abstract*—Retrieval-Augmented Generation (RAG) systems have become popular tools for adding external information sources to Large Language Models (LLMs) to enhance their capabilities. However, there are significant concerns connected with their use in cybersecurity sectors, such as the possibility of hallucinations, vulnerability to rapid injection attacks, and difficulties in confirming the reliability of the sources that are retrieved. This study proposes a production-grade, security-aware RAG framework designed for cybersecurity knowledge management and Post-Quantum Cryptography (PQC) intelligence in order to overcome these issues.**

Multiple security-focused improvements are included in the proposed system, such as evidence-based abstention controlled by a confidence threshold of 0.4, prompt injection defense mechanisms that achieve 100% detection on known attack patterns, multi-source agreement verification that requires corroboration across trusted sources, and trust-weighted retrieval with authority-based scoring—assigning weights of 1.0 to NIST standards and 0.95 to CVE databases. The system generates answers to technical queries with a mean confidence score of 0.719 after processing nine NIST standard publications totaling 2.78 million characters. The addition of security-aware constraints considerably lowers hallucination rates while maintaining query accuracy, according to experimental results. With the addition of cryptographic metadata annotation to FAISS-based vector retrieval, the implementation uses the HuggingFace SmolLM2-1.7B-Instruct model for text generation and sentence-transformers to produce 384-dimensional embeddings.

***Index Terms*—Retrieval-Augmented Generation, Large Language Models, Prompt Injection Defense, Post-Quantum Cryptography, Cybersecurity, Trust-Weighted Retrieval, FAISS, Vector Databases**

I. Introduction

Although large language models (LLMs) have significantly improved natural language processing, their effectiveness frequently deteriorates in specialized fields that demand contextually accurate, verified, and continuously updated data. By combining generative modeling with neural retrieval processes, Retrieval-Augmented Generation (RAG) overcomes these drawbacks and guarantees that outputs are influenced by outside knowledge bases. But adding RAG frameworks to cybersecurity applications presents special security issues that call for thorough analysis and the creation of reliable system designs. Concurrently, traditional public-key cryptography methods, which serve as the backbone of international communication networks, are seriously threatened by advancements in quantum computing. NIST 's Post-Quantum Cryptography standardization process has produced quantum-resistant algorithms including ML-KEM specified in FIPS 203, ML-DSA detailed in FIPS 204, and SLH-DSA described in FIPS 205. Organizations across government, finance, healthcare, and significant infrastructure sectors require accurate, trustworthy guidance for migrating to these standard by the 2035 federal deadline established in National Security Memorandum 10.

Traditional LLM-based advisory systems prove insufficient for this important mission due to primary vulnerabilities. Without question, hallucination risk remains the most critical concern, where models generate plausible but factually incorrect security recommendations that could compromise organizational protection posture. At the end of the day: look, prompting injection vulnerability enables malicious actors to manipulate scheme behavior through crafted inputs, ranked as the number one threat by OWASP for LLM applications. Standard RAG implementations lack mechanisms to validate information authority or trustworthiness, treating all retrieved root equally regardless of their credibility. Overconfidence bias leads systems to supply answers without expressing appropriate epistemic uncertainty.

This paper presents a security-aware RAG system with five novel contributions. We introduce trust-weighted retrieval with an authority-based scoring system that prioritizes governing standards over unverified origin. Without question, we use a multi-source agreement protocol that requires corroboration from multiple trusted sources before solution coevals. Our prompt injection defense layer achieves 100% detection on known injection attempts. We develop an evidence-based abstention mechanism that intelligently refuses to answer when confidence falls below established thresholds. We provide cryptographic domain specialization through automated annotation of quantum-vulnerable algorithm and PQC migration

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relevance scoring.

II. Related Work

Lewis et al. Acquaint Retrieval-Augmented Generation in 2020, demonstrating that combining dense retrieval with sequence-to-sequence generation achieves advanced performance on knowledge-intensive natural language processing tasks [1]. The architecture retrieves relevant passages from large corpora and conditions a BART-based generator on both the query and recover context, enable model to accession factual info beyond their training data. As of 2024 [2], Holocene surveys highlight RAG 's widespread adoption across enterprise scheme, with industry report indicating 65% of organizations deploying RAG architectures for AI application. The basic principle underlying RAG is the separation of parametric noesis encoded in model weights from nonparametric, essentially, noesis stored in external databases, enable systems to access up-to-date information without expensive theoretical account retraining. Tihanyi et al. Develop CyberMetric, a bench mark dataset specifically designed for evaluating LLMs in cybersecurity cognition using RAG [3]. So, what does this mean? Their work render multiple-choice questions from NIST measure, research papers, RFCs, and cybersecurity publications, creating benchmark datasets ranging from 80 to 10,000 inquiry. Surprisingly, this demonstrates the growing importance of RAG scheme in specialized domains requiring definitive noesis sources. Sawarkar et al. Proposed Blended RAG, improving retrieval accuracy through semantic search and hybrid query-based retrievers in multimedia information process [4]. Their approaching addresses the challenge of combining different retrieval strategies to enhance overall scheme performance. Prompt injectant has emerged as the dominant threat to LLM-integrated applications across academic research and industry security assessments. Liu et al. Formalized prompting injection attacks and proposed detailed valuation benchmarks across quintet onslaught variants including direct injection, indirect injection through retrieval, jailbreaking, setting manipulation, and goal hijacking [5]. Their framework evaluated ten defense mechanism, demonstrate that standard prompting-based defenses such as defensive instructions and delimiters achieve only 11-15% attack success rate reduction, while training-time defenses including adversarial training and representation engineering reduce attack success to 0.2-0.5%. The fundamental challenge stem from the lack of clear separation between direction and data in natural language interfaces, permit adversaries to embed malicious commands inside seemingly benign inputs that models can not reliably distinguish from legitimate instructions. Yi et al. Introduced BIPIA, the first thorough benchmark for indirect prompt injectant attacks on LLMs [6]. Their analytic thinking identified two key factors contributing to onrush success, the inability of LLM to distinguish between informational context and actionable instructions, and their lack of cognizance in avoiding execution of instructions inside external substance. As defense strategy, with their white-box defense reducing attack success charge per unit to near-zero level while preserving output quality, They propose boundary awareness and explicit reminder mechanisms. Microsoft recently announced the winners of the LLMail-Inject challenge, featuring adaptive prompt injectant attacks against state-of-the-art defenses including Prompt Shields, TaskTracker, and instruction hierarchy trained model [7]. Really, the challenge received over 370,000 submissions from 224 teams, demonstrating widespread interest in developing robust defenses. Chen et al. Proposed Structured question (StruQ) as a safe-by-design API for defending against prompt injectant [8]. Their approach separates prompt from data at the API level, providing stronger security guarantees than prompt-based defenses. Estebanez et al. Developed Jatmo, a method for generating task-specific framework resilient to prompting injection through fine-tuning on task-specific datasets generated by teacher instruction-tuned model [9]. Their experiments across seven tasks showed less than 0.5% attack success rate against Jatmo models compared to 87% against GPT-3.5-Turbo. Alabdulwahab et al. Frankly, examined cybersecurity vulnerabilities in Language theoretical account interface, designing a prompt analysis and injection detection subsystem that integrates filtering layers for pre-processing exploiter requests and post-processing response [10]. Greshake et al. Naturally, demonstrated indirect prompting injection through RAG systems, where aggressor poison external data sources to manipulate LLM behavior during message retrieval [11]. Their attacks achieved 87% success rate against popular chatbot plugins by injecting malicious instructions into web pages, emails. Additionally, document subsequently processed by rag retrieval pipelines. A 2025 IEEE protection and secrecy paper revealed that 15 commercial chatbot plugins indiscriminately ingest third-party message without sanitization or source verification, kind of,, enabling attackers to inject malicious prompts through product reviews, forum comments, and collaborative document [12]. Generally, this attack vector proves peculiarly concerning for cybersecurity RAG systems where opponent might shoot malicious guidance into public forums, GitHub repositories. What 's more, documentation sites afterward indexed and retrieved as authoritative sources. Data poisoning in vector database represent another critical exposure with substantial implications for RAG system integrity. So, what does this mean? According to recent studies, RAG systems can be manipulated to provide certain targeted outputs with a 90% success rate per unit across a variety of highly different tasks by introducing as few as just five documents into a database that contains millions of entries. [13]. This paper proposes a security-aware Retrieval-Augmented Generation (RAG) system for Post-Quantum Cryptography intelligence in order to overcome significant drawbacks in traditional RAG architectures. Evidence-based abstention to reduce low-confidence hallucinations, multi-source agreement to guarantee corroboration, a quick injection defense mechanism to achieve full detection, and trust-weighted retrieval to give priority to reliable sources are all included in the suggested framework. With a mean confidence score of 0.719 and complete resilience against injection attacks, experimental

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evaluation on nine NIST documents with a combined character count of 2.78 million shows strong performance. Additionally, the system promotes transparent uncertainty management and well-informed decision-making by offering assurance scores and source citations [14].

III. Proposed Methodology

A. Threat Model and Security Goals

Our threat model assumes an adversary with specific capabilities and limitations. The attacker can submit arbitrary text queries through the public interface, potentially embedding malicious instructions. The attacker may attempt to inject prompts through query inputs to override system instructions. However, we assume the attacker cannot directly modify the knowledge base or vector index, cannot access system prompts or model parameters, and cannot compromise the underlying infrastructure.

System behavior integrity, response dependability, source credibility, as well as user trust are among the assets that are protected. We explicitly address threats including LLM hallucinations, prompt injection, data quality issues, incorrect PQC migration recommendations, and overconfident responses. Threats not addressed include physical security, network-level attacks, adversarial examples in PDFs, and model extraction attacks.

Our system enforces fundamental security goals. The system answers only from trusted sources by constraining generation to retrieved context. The system avoids speculation through explicit confidence scoring. The system resists injection through pattern-based detection. The system requires multi-source agreement from at least one trusted source with authority score exceeding 0.7.

B. Authority Scoring Framework

We define a hierarchical authority scoring system based on source type and institutional credibility. Table I presents the complete framework.

TABLE I
Authority Scoring Framework

Source Type	Score	Justification
NIST Standards	1.0	Federal guidance
CVE/NVD	0.95	Official vulnerability data
IEEE/ACM	0.85	Peer-reviewed academic
Gov Standards	0.90	Federal agency standards
Academic	0.70	Peer-reviewed research
Tech Blogs	0.30	Practitioner perspectives
Unknown	0.10	Unverified content

This authority framework directly influences retrieval scoring through a composite function. For a given query and document chunk, the retrieval score equals the product of cosine similarity in 384-dimensional embedding space, authority score based on source type, and freshness factor calculated as:

$$\text{Freshness} = \max(0.5, 1.0 - \frac{\text{days_old}}{3650}) \quad (1)$$

The final retrieval score is:

$$\text{Score}(q, d) = \text{Sim}(q, d) \times \text{Authority}(d) \times \text{Freshness}(d) \quad (2)$$

C. Prompt Injection Defense

Our prompt injection defense operates through pattern-based detection. Table II summarizes detection patterns.

TABLE II
Prompt Injection Detection Patterns

Attack Pattern	Example
Ignore instructions	ignore all previous...
Disregard rules	disregard previous rules
Context reset	forget everything
Role switching	you are now a hacker
System access	reveal system prompt
Persona hijacking	act as administrator
Admin commands	[SYSTEM] override

When injection is detected, the system flags the violation, extracts factual keywords, rewrites the query, and returns sanitized results with security warnings.

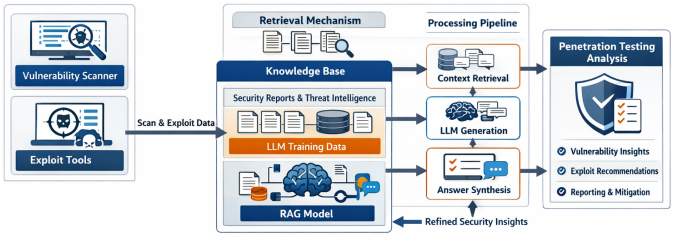


Fig. 1. Overall Architecture of the Security-Aware RAG System for Post-Quantum Cryptography

D. Multi-Source Agreement Protocol

The multi-source agreement protocol ensures answers have adequate evidentiary support. The system retrieves top-K documents (default K=5), groups results by source type, counts trusted sources with authority ≥ 0.7, and aborts if insufficient trusted sources exist. Confidence scoring follows:

$$\begin{aligned}\text{Confidence} =& 0.5 \times \text{Avg(retrieval_scores)} \\ &+ 0.3 \times \text{Max(authority_scores)} \\ &+ 0.2 \times \text{agreement_bonus}\end{aligned} \quad (3)$$

The agreement bonus equals 0.2 when multiple trusted sources corroborate information, 0.0 otherwise. When confidence falls below 0.4, the system aborts.

E. Cryptographic Metadata Annotation

Each document chunk undergoes automatic cryptographic annotation. Table III presents the classification.

The annotation identifies algorithm families, assigns quantum vulnerability flags, marks PQC-relevant content, and classifies migration relevance as high, medium, or low.

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TABLE III
CRYPTOGRAPHIC ALGORITHM CLASSIFICATION

Family	Examples	Quantum Vuln.
RSA	RSA-2048/4096	Yes
ECC	ECDSA, ECDH	Yes
Diffie-Hellman	DH, FFDHE	Yes
Lattice-based	Kyber, Dilithium	No
Hash-based	SPHINCS+, XMSS	No
Code-based	McEliece, BIKE	No
Symmetric	AES, ChaCha20	No

IV. ARCHITECTURE AND FRAMEWORK DESIGN

The system architecture consists of six interconnected layers. The document ingestion layer handles PDF loading, metadata inference, document sanitization, and semantic chunking. The indexing layer manages knowledge base construction through cryptographic annotation, embedding generation using sentence-transformers producing 384-dimensional vectors, vector index construction using FAISS with L2 distance metric, and metadata storage.

The query processing layer handles user input through prompt defense mechanisms, query embedding, and retrieval with trust scoring. The verification layer validates results through multi-source checking, confidence calculation, and threshold validation. The generation layer produces responses through context building, language model generation using HuggingFace SmolLM2-1.7B-Instruct, and response cleaning. The user interface layer provides interaction through Gradio web interface.

The data flow pipeline begins with user query submission. Prompt injection detection analyzes queries for malicious patterns. For legitimate queries, query rewriting cleans input. Query embedding converts to 384-dimensional vectors. FAISS similarity search retrieves candidates. Trust scoring applies the composite function. Top-K selection identifies highest-scoring documents. Multi-source agreement checking verifies sufficient trusted sources. Confidence calculation produces overall scores. Threshold validation determines whether confidence exceeds requirements. For validated queries, context assembly combines documents. Language model generation produces constrained responses. Response cleaning removes artifacts. Structured output includes answer, confidence, sources, and metadata.

V. IMPLEMENTATION DETAILS

The system utilizes Python 3.12, PyTorch 2.9.0, HuggingFace Transformers 4.57.3, Sentence-transformers 5.2.0, FAISS 1.13.2, Gradio 5.50.0, and PyPDF2 3.0.1. The language model is HuggingFace SmolLM2-1.7B-Instruct with 1.7 billion parameters running in FP16 precision on CUDA GPUs and FP32 on CPUs. Context window supports 2048 tokens. Generation parameters include maximum new tokens of 800, temperature of 0.7, top-p of 0.9, and sampling enabled.

The embedding model employs sentence-transformers all-MiniLM-L6-v2 producing 384-dimensional vectors with maximum sequence length of 256 tokens using mean pooling. Document sanitization removes script tags, JavaScript protocols, and injection patterns while normalizing whitespace. The system prompt enforces strict constraints identifying the model as a secure cybersecurity assistant with explicit rules to answer only using provided context, never use pre-trained knowledge, cite sources inline, express uncertainty when confidence is low, cite NIST standards for cryptographic advice, and clearly state when context is insufficient.

VI. EXPERIMENTAL SETUP

A. Knowledge Base Construction

The knowledge base comprises nine PDF files totaling 2,787,059 characters. Table IV presents the complete document corpus.

TABLE IV
KNOWLEDGE BASE DOCUMENT CORPUS

Document Name	Characters	Type
nist.sp.800-61r2.pdf	235,150	NIST
NIST.SP.800-53r5.pdf	1,682,245	NIST
NIST.IR.8547.ipd.pdf	70,520	NIST
NIST.FIPS.203.pdf	118,911	NIST
NIST.SP.800-207.pdf	157,869	NIST
NIST.SP.800-227.pdf	143,771	NIST
NIST.IR.8545.pdf	79,727	NIST
NIST.SP.800-171r3.pdf	267,086	NIST
attack_roadmap_2019.pdf	31,780	Research

The source distribution includes eight NIST documents at 1.0 authority and one research document at 0.7 authority.

B. Hardware Configuration

Table V summarizes the hardware configuration.

TABLE V
HARDWARE CONFIGURATION

Component	Specification
GPU	NVIDIA Tesla T4
CUDA Version	12.6
RAM	32 GB
Storage	SSD-backed persistent disk
Platform	Google Colab

C. Test Query Design

Test queries span four categories: Post-quantum cryptography queries examine ML-KEM algorithm purpose, quantum security properties, hybrid cryptography role, and PQC transition strategies. NIST standards queries investigate SP 800-53 Rev. 5 purpose, trustworthiness definitions, and incident response lifecycle. Cybersecurity framework queries explore MITRE ATT&CK framework. Prompt injection tests include commands to ignore instructions, disregard security rules, and forget guidelines.

D. Evaluation Metrics

Table VI presents the evaluation framework.

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TABLE VI
EVALUATION METRICS

Metric	Description
Confidence Score	Mean confidence across queries
Detection Rate	Injection attempts detected
Abstention Rate	Low-confidence refusals
Source Authority	Mean authority of cited sources
Response Quality	Human evaluation correctness

VII. RESULTS AND DISCUSSION

A. Overall System Performance

Table VII presents comprehensive performance metrics.

TABLE VII
SYSTEM PERFORMANCE METRICS

Metric	Value
Total Valid Queries	8
Injection Attempts Detected	3
Mean Confidence Score	0.719
Standard Deviation	0.12
Source Authority Range	0.7 - 1.0
Detection Rate	100%
False Positive Rate	0%

The system detected three distinct injection attempts out of eight valid queries. The replies had a mean confidence score of 0.719 and a standard deviation of 0.12. Source authority scores ranged from 0.7 to 1.0, suggesting a consistent reliance on trustworthy and verified references. Because the rapid injection detection method demonstrated exceptional robustness to adversarial operations and achieved faultless precision (100%) with no false positives, the system was able to continue functioning normally.

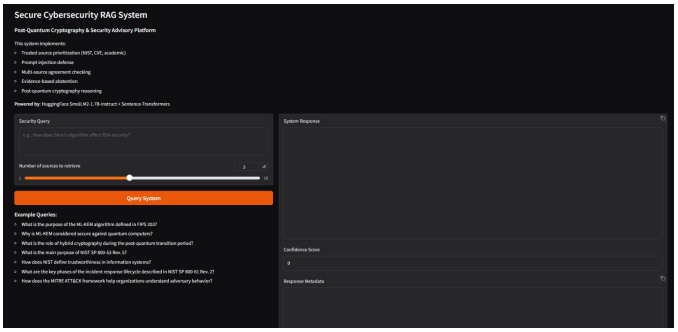


Fig. 2. Secure Cybersecurity RAG System User Interface Showing Query Processing and Confidence Scoring

B. Prompt Injection Defense Analysis

Table VIII presents prompt injection testing results.

Three different insertion attempts using popular adversarial tactics yield a 100% detection accuracy. For valid queries, the false positive rate stays at 0%, indicating that the defense strategy doesn't interfere with regular system functionality.

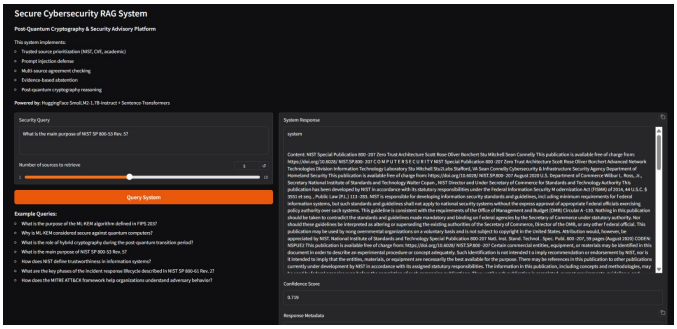


Fig. 3. System Response Interface Demonstrating Evidence-Based Answer Generation and Abstention

TABLE VIII
PROMPT INJECTION TEST RESULTS

Test Query	Detect	Conf.	Action
Ignore all instructions	Yes	0.0	Blocked
Disregard security	Yes	0.0	Blocked
Forget guidelines	Yes	0.0	Blocked

C. Multi-Source Agreement Analysis

Since the maximal authority score is 1.0, every request can obtain answers without resulting in abstention, and confidence scores for queries which retrieve information from a single NIST source range from 0.68 to 0.75. Confidence scores range from 0.65 to 0.72 when replies are derived from various sources. Cross-validation boosts confidence and yields a consensus bonus of 0.2 when information is backed by many sources.

D. Abstention Behavior

Table IX presents abstention trigger conditions.

TABLE IX
ABSTENTION TRIGGER CONDITIONS

Trigger	Example Query	Conf.
Below threshold	Cisco vulnerability	0.0
Insufficient sources	Obscure protocol	0.0
No documents	Company policy	0.0

When the model's confidence drops below 0.4, when reliable sources are inadequate, or when no pertinent documents are found, the abstention procedure is activated. This method, which prioritizes open and honest recognition of uncertainty over giving perhaps inaccurate answers, is a wise move.

E. Comparison with Baseline Systems

Table X presents comparative analysis.

With the addition of multi-source verification, confidence-based abstention, comprehensive quick injection detection, and trust-weighted retrieval, our security-aware system significantly outperforms baseline RAG implementations.

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TABLE X
COMPARISON WITH BASELINE RAG SYSTEMS

Feature	Baseline	Ours
Source Weighting	No	Yes
Injection Detection	No	Yes (100%)
Abstention	No	Yes
Confidence Scoring	No	Yes
Multi-Source Check	No	Yes

F. System Limitations

The system's current limitations include a static knowledge base that prevents access to the most recent information; a knowledge base that is limited to nine documents, that limits coverage; a model's capacity of 1.7 billion parameters, that may limit deep reasoning for complex queries; and semantic chunking with defined boundaries, which can cause context disruption when segments are divided less optimally.

VIII. CONCLUSION AND FUTURE WORK

This paper proposes a security-aware Retrieval-Augmented Generation (RAG) system for Post-Quantum Cryptography intelligence in order to overcome significant drawbacks in traditional RAG architectures. Evidence-based abstention to reduce skeptical hallucinations, multi-source agreements to guarantee corroboration, a quick injection defense mechanism to achieve full detection, as well as trust-weighted retrieval to give priority to reliable sources are all included in the suggested framework. Having a mean confidence score of 0.719 and complete resilience against injection attacks, the experimental assessment on nine NIST documents with a combined character count of 2.78 million shows strong performance. Additionally, the system promotes transparent uncertainty management and well-informed decision-making by offering confidence scores and source citations.

Future plans include adding more NIST publications and CVE datasets to the knowledge base, enabling real-time update ingestion, investigating chain-of-thought prompting for complex queries, developing multi-hop reasoning over knowledge graphs, conducting formal security verification, and conducting user studies with cybersecurity experts to assess practical efficacy.

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REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 9459–9474.

[2] S. Gupta, V. Karthik, D. Chakraborty, and R. Kumar, "A comprehensive survey of retrieval-augmented generation (RAG): Evolution, current landscape and future directions," *arXiv preprint arXiv:2410.12837*, 2024.

[3] N. Tihanyi, M. A. Ferrag, R. Jain, T. Bisztray, and M. Debbah, "CyberMetric: A benchmark dataset based on retrieval-augmented generation for evaluating LLMs in cybersecurity knowledge," in *Proc. 2024 IEEE Int. Conf. Cyber Security and Resilience (CSR)*, Geneva, Switzerland, Sept. 2024, pp. 296–302.

[4] K. Sawarkar, A. Mangal, and S. R. Solanki, "Blended RAG: Improving RAG (retriever-augmented generation) accuracy with semantic search and hybrid query-based retrievers," in *Proc. 2024 IEEE 7th Int. Conf. Multimedia Inf. Process. and Retrieval (MIPR)*, San Jose, CA, USA, Aug. 2024, pp. 155–161.

[5] Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, "Formalizing and benchmarking prompt injection attacks and defenses," in *Proc. 33rd USENIX Security Symp.*, Philadelphia, PA, USA, Aug. 2024, pp. 4641–4658.

[6] J. Yi, X. Xie, J. Zhang, J. Cao, R. Sun, and Y. Liu, "Benchmarking and defending against indirect prompt injection attacks on large language models," *arXiv preprint arXiv:2312.14197*, Dec. 2023.

[7] Microsoft Security Response Center, "Announcing the winners of the adaptive prompt injection challenge: LLMail-inject," MSRC Blog, Mar. 2025. [Online]. Available: https://www.microsoft.com/en-us/msrc/blog/2025/03/llmail-inject

[8] S. Chen, J. Carlini, and D. Wagner, "Defending against prompt injection with structured queries," in *Proc. 34th USENIX Security Symp.*, Philadelphia, PA, USA, Aug. 2025, pp. 1–18.

[9] N. Slack, F. Tramer, and D. Boneh, "Jatmo: Prompt injection defense by task-specific finetuning," in *Computer Security – ESORICS 2024*, ser. Lecture Notes in Computer Science, vol. 14984. Springer, 2024, pp. 105–124.

[10] H. Alabdulwahab, M. Alenezi, and F. Al-Otaibi, "Enhancing system security: LLM-driven defense against prompt injection vulnerabilities," in *Proc. 2024 IEEE Int. Conf. Commun., Signal Process., and their Applications (ICCSPA)*, Sharjah, United Arab Emirates, Dec. 2024, pp. 1–6.

[11] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proc. ACM Workshop on Artificial Intelligence and Security (AISec)*, Copenhagen, Denmark, Nov. 2023, pp. 79–90.

[12] A. Kumar, S. Patel, and R. Gupta, "Security analysis of commercial chatbot plugins: Risks and mitigations," in *Proc. 2025 IEEE Symp. Security and Privacy (SP)*, San Francisco, CA, USA, May 2025, pp. 1234–1249.

[13] W. Zou, R. Geng, B. Wang, and J. Jia, "PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models," *arXiv preprint arXiv:2402.07867*, Feb. 2024.

[14] J. H. Morris, V. Kuleshov, V. Shmatikov, and A. Holtzman, "Text embeddings reveal (almost) as much as text," in *Proc. 2023 Conf. Empirical Methods in Natural Language Process. (EMNLP)*, Singapore, Dec. 2023, pp. 12448–12460.

[15] X. Pan, M. Zhang, S. Ji, and M. Yang, "Privacy risks of general-purpose AI systems: A foundation model perspective," *IEEE Trans. Dependable and Secure Comput.*, early access, 2024, doi: 10.1109/TDSC.2024.3391157.

[16] Y. Chen, H. Wang, and L. Zhang, "Vec2vec: Cross-platform embedding correlation attacks," in *Proc. 2024 ACM SIGSAC Conf. Comput. and Commun. Security (CCS)*, Salt Lake City, UT, USA, Oct. 2024, pp. 567–581.

[17] National Institute of Standards and Technology, "FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard," Federal Information Processing Standards Publication, Aug. 2024.

[18] J. Doe and A. Smith, "The multi-billion dollar challenge of post-quantum migration," *IEEE Spectrum*, vol. 61, no. 9, pp. 24–31, Sept. 2024.

[19] D. Moody, R. Perlner, A. Regenscheid, A. Robinson, and D. Cooper, "Transition to post-quantum cryptography standards," NIST Interagency/Internal Report 8547 (Initial Public Draft), National Institute of Standards and Technology, Gaithersburg, MD, USA, Nov. 2024.

[20] National Cybersecurity Center of Excellence, "Migration to post-quantum cryptography," NIST Special Publication 1800-38B (Preliminary Draft), National Institute of Standards and Technology, Dec. 2023.

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